

Lab: Bio-Inspired System Architecture Design

Objective

Analyze biological organisms as Active Inference systems and design bio-inspired robotic architectures. You will compare the Markov blanket structure of biological systems (insects, mammals, cephalopods) with robotic systems and propose design principles that transfer biological organizational patterns to robotic hardware.

Prerequisites

- Completion of the corresponding module.md lecture material
- Familiarity with Active Inference concepts (generative models, free energy, prediction errors)
- Basic pseudocode and diagram skills

Part 1: Biological System Analysis

Select a biological organism (e.g., ant, octopus, honeybee) and identify its system architecture:

1. Map the organism's sensory systems (eyes, antennae, mechanoreceptors) to robotic sensor equivalents.
2. Map the organism's motor systems (muscles, cilia, jet propulsion) to robotic actuator equivalents.
3. Identify the Markov blanket: what constitutes internal states (nervous system), sensory states, active states, and external states?
4. How does the organism's body plan (segmented, radial, bilateral) affect its generative model structure?

Write your response here

Part 2: Biomimetic Design Transfer

For your chosen organism, design a robotic system that mimics its architecture:

1. Sensor placement: How does the biological sensor layout (e.g., compound eyes, distributed mechanoreceptors) inspire your robot's sensor configuration?
2. Actuator design: What biological motor principles (antagonistic muscles, hydrostatic skeletons, peristalsis) could improve your robot's actuation?
3. Nervous system architecture: How does the organism's neural organization (centralized brain vs. distributed ganglia) inform your computational architecture?

Write your response here

Part 3: Scaling Laws and Constraints

Biological systems obey physical scaling laws that constrain their design. Analyze:

1. How does body size affect sensor resolution, actuator force, and neural processing time in biology?
2. What scaling challenges arise when transferring insect-scale designs to human-scale robots (or vice versa)?
3. How does the free energy principle explain why organisms at different scales converge on similar architectural patterns?

Write your response here

Part 4: Design Challenge Comparison

Compare three bio-inspired architectures for a search-and-rescue robot:

Feature	Insect-Inspired	Mammal-Inspired	Octopus-Inspired
Body plan			
Sensor strategy			
Locomotion			
Control architecture			
Generative model depth			

Write your response here

Summary Table

Concept	Classical Robotics	Active Inference	Your Design
Core mechanism			
Uncertainty handling			
Adaptation			
Key advantage			

References

- Lanillos, P., et al. (2021). Active Inference in Robotics and Artificial Agents. *Frontiers in Neurorobotics*.
- Parr, T., Pezzulo, G., & Friston, K. J. (2022). *Active Inference*. MIT Press.
- Pio-Lopez, L., Nizard, A., Friston, K., & Pezzulo, G. (2016). Active Inference and robot control. *Journal of the Royal Society Interface*, 13(122).